Workforce Analytics | Executive Brief

Decision brief

DERIVED FROM SIMULATED DATA: 1,470 HR records include 237 Attrition Yes labels (16.1%); these are not annual turnover rates.

Highest observed fictional role share: Sales Representative, 33/83 (39.8%). Small group support and role composition matter.

Held-out HR logistic at validation-selected threshold 0.32: precision 0.500, recall 0.511, F1 0.505, AP 0.572, ROC AUC 0.785; test n=294.

Independent Adult income benchmark test n=16,281; random_forest, threshold 0.34, F1 0.693, AP 0.790. Positive means >$50K historical income.

Recommended next steps

Investigate workload and role design through voluntary employee feedback; fictional correlations cannot justify policy changes on their own. In a real organization, validate definitions, establish dated hire/exit histories, and measure retention during an approved pilot.

Do not use models for hiring, dismissal, pay setting or employee surveillance. Aggregate support planning requires contemporary representative data, employee consultation and independent governance.

Track cohort retention, overtime prevalence, exit reasons, prediction calibration, subgroup error rates and missingness after collecting real data. Set thresholds with stakeholder-defined costs, not this educational F1 optimum.

Boundaries

All HR findings derive from fictional records. BLS median tenure 4.1 years in January 2026 is context, not a performance target or like-for-like comparison. No savings, discrimination or future attrition prediction is established.

Sources and data license

Fictional IBM HR data; ODbL 1.0 / DbCL 1.0. Derived educational analysis, not employee decisions.

IBM HR source and license: https://github.com/IBM/employee-attrition-aif360#dataset-license-details

Original fictional HR listing: https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset

Adult: Becker & Kohavi (1996), DOI 10.24432/C5XW20: https://archive.ics.uci.edu/dataset/2/adult

BLS Employee Tenure in 2026, released September 24 2026: https://www.bls.gov/news.release/tenure.nr0.htm

Data terms: ODbL 1.0 and DbCL 1.0; full license links and alteration methods are in NOTICE.md. Adult: CC BY 4.0. BLS statistics: public domain.

